INTRODUCTION: -

1.1 OVERVIEW

AI has come a long way from being a command-response system to an advanced conversational agent. But traditional chatbots based on text are still restricted from delivering an emotionally engaging and visually interactive experience, just like human communication. The conversational AI field has started to overcome these challenges by introducing multimodal integration.

These are exciting developments, but existing solutions suffer from a number of issues at present: Cloud-based AI services, poor expressiveness of the avatar and communication/layering latency in multimodal interaction frameworks that are not integrated into the system. As of today, they are mainly focusing on one aspect, either conversational intelligence or visual, but not on a real-time environment with true integration of both.

This report presents the end-to-end real-time AI avatar chatbot system KARS AI to solve the above problems, which includes innovative local inference of large language models, real-time 3D avatar animation, and multimodal communication protocol. Built using React.js/Vite framework, with FastAPI backend services, Supabase serves as the cloud infrastructure, and Ollama is used for processing locally to enable emotionally expressive digital interactions.

1.2 MOTIVATION

The primary motivation driving the development of KARS AI stems from the stark limitations inherent in contemporary digital assistants. While current AI models can generate highly accurate text, they frequently deliver a sterile and fragmented user experience. Existing avatar solutions in the market often suffer from three major bottlenecks: severe communication latency during multimodal interactions, poor visual expressiveness, and an over-reliance on cloud-based AI services, which raises significant data privacy and security concerns for end-users.

There is a compelling need for a holistic system that seamlessly bridges the gap between high-level conversational AI and realistic visual representation without compromising user privacy or system performance. The motivation behind KARS AI is to engineer an emotionally intelligent digital companion that operates in real-time, targeting sub-second bidirectional communication latency (under 950ms).

By transitioning the heavy cognitive reasoning to a local environment via Ollama, the project aims to eliminate the privacy risks and latency bottlenecks associated with third-party cloud LLMs. Ultimately, KARS AI is driven by the ambition to redefine human-machine interaction, moving beyond simple transactional utility to foster genuine, empathetic engagement in critical fields spanning from remote education and customer support to virtual companionship and digital mental health therapy.

2 PROJECT SCOPE: -

2.1 System Boundaries and Core Objectives:

The primary scope of this project encompasses the end-to-end design, development, and deployment of "KARS AI," a comprehensive, real-time multimodal avatar system. The system boundary is defined by its objective to deliver an emotionally rich, immersive human-computer interaction platform that significantly outperforms traditional text-based conversational agents. The project focuses on creating a seamless, modular pipeline that captures user input (voice or text), processes cognitive logic using localized artificial intelligence, and outputs a synchronized audiovisual response within a sub-second latency threshold.

2.2 Technical and Architectural Scope:

The technical scope of KARS AI is strictly defined by a modern, distributed full-stack architecture designed for scalability, high performance, and user privacy:

- **Frontend Scope:** The user interface is restricted to a web-based environment built using React.js and Vite. It includes the integration of a 3D rendering engine utilizing Three.js and React Three Fiber to handle the visual representation of the digital persona.
- **Backend and Communication Scope:** The backend orchestration is handled by FastAPI, ensuring high-performance RESTful API routing. A critical component of the scope is the implementation of WebSockets to maintain persistent, bidirectional, low-latency communication between the client and server.
- **Cognitive Engine Scope:** Unlike traditional systems that rely entirely on cloud-based AI, this project's scope explicitly includes the integration of Ollama as a local Large Language Model (LLM) inference engine. This ensures offline cognitive capabilities, zero API cost overheads, and stringent data privacy.
- **Data Persistence Scope:** The cloud infrastructure is scoped to use Supabase, handling secure user authentication, real-time database synchronization, and persistent chat history management.

2.3 Multimodal and Functional Scope:

The functional scope of KARS AI extends beyond basic dialogue, encompassing several advanced multimodal capabilities:

- **Audiovisual Synchronization:** The project includes the development of real-time lip-syncing algorithms. By analyzing audio waveforms from the Text-to-Speech (TTS) module, the system dynamically manipulates the 3D avatar's facial blend shapes (visemes) and overlays contextual emotional expressions and hand gestures.
- **Specialized Application Modes:** The scope incorporates advanced functionality tailored for specific use cases. This includes an emotionally intelligent Therapy Mode designed with empathetic conversation templates, and an AI Schedule Planner to assist users with real-world task organization.
- **State Management:** The avatar is scoped to possess dynamic behavioral states—specifically categorized as Listening, Talking, and Explaining—which adapt in real-time based on the flow of the conversation.

LITERATURE SURVEY

A. The evolution of conversational AI systems.

AI-powered chatbots have developed into much more than rule-based systems; they are now advanced multimodal chatbots. Initial studies were mainly on text-based interactions but in recent years, the role of embodied conversational agents has been brought to the fore. To enable human-level casual conversation in virtual reality, [T. Yamazaki et al., 2023] proposed an open domain avatar chatbot with multimodal capabilities using the large language model. They found a number of problems, including their slow generation and the challenge of matching the different types of output with the different avatar control systems.

B. Integrating multimodal AI and avatar technology. Connecting multimodal AI and avatar.

The last few years, the study of multi-modal interaction has gained increasing interest, making it a goal for more natural user experiences. E. Aliyev et al., (2025) gave an extensive overview of conversational avatars powered by AI, focusing on the convergence of text-to-speech, speech recognition, and open-domain dialogue systems to construct responsive and realistic avatars for conversation. Their analysis showed that multimodal systems offer a substantial boost to the adoption of AI systems and improve the trustworthiness of the systems compared with single-modal systems.

Abdul Basit & Muhammad Shafique introduced tinyDigiClones framework that tackles computational challenges in edge-optimized personalized avatars. They showed that using cutting-edge LLMs on resource-constrained devices with minimal latency involves careful optimization strategies and the selection of light models.

C. An overview of emotion-aware and empathetic systems.

One of the most important research trends has been the incorporation of emotional intelligence into conversational Artificial Intelligence (AI). Shine P Xavier & Saju P John [2024] proposed the empathetic chatbot system for real-time multimodal fusion that understands the emotions expressed in a user's voice, written word and image. Unlike the current facial emotion recognition systems that rely on a single modality,

they adopted EmbraceNet framework and Botox Optimization based CNNs to achieve better performance in facial emotion recognition.

V. Devi et al., 2024, proposed a framework, called EMPATHIC that combines real-time face recognition and emotion detection with natural language processing in the context of emotionally aware multimodal chatbots. They highlighted the importance of training quality data for dialogue performance and introduced a set of dialogue instruction templates based on vision and language data.

D. Educational and Specialized Applications

Avatar-based AI systems have demonstrated the promising potential of educational application. In [Arnav Atkare et al., 2025] they built a full-body digital human using low-latency audio processing and Gemini for reasoning, and ElevenLabs for speech synthesis, for real-time conversational learning. Their assessment based on cognitive load theory principles proved to be successful in achieving the expected results: higher learner retention and satisfaction while the median time taken was still sub-2 second.

With the help of VRoid's ability to design 3D avatars and Meta Llama 3.8B's natural language processing abilities, it was possible to show an adaptive, AI-powered educational assistant which can offer instant explanation on different topics and create interactive learning experiences.

E. Technical Implementation Approaches

Several technical approaches have been explored to use avatars in conversation systems. According to the research conducted by Jayrajsinh Gohil et al., in 2025, there is great potential in using Ollama and LLama3 in university chatbots due to local installation of LLMs. In the work [Sasirekka R et al., 2026] they introduced a multi-client video conferencing application-based system to enable real-time video conferencing sessions with AI agents, which uses WebRTC for connection maintenance and integrated prompt engineering to adapt the role of the AI agent.

F. Research gaps and limitations

Although notable progress has been made, current research shows that current solutions either lack a proper integration between conversational AI and visual representation or focus on either one without the other, or only address limited aspects

of the problem, such as avatar expressiveness and emotional response, or are missing a holistic solution of integrating local LLM inference with real-time multimodal interaction. These gaps underscore the importance of a system that can tackle multiple problems simultaneously, is scalable, and performs well, such as KARS AI.

PROBLEM ANALYSIS: -

4.1 PROBLEM DEFINITION

As AI has evolved, it has taken systems from command and response to smart, conversational agents. But conventional chatbots, which are still rule-based and rely on text input, are still limited in their ability to provide users with a more engaging and visually stimulating experience, as in natural human communication. Though some preliminary work has been performed in the integration of multimodal, there are serious systematic problems with the current solutions based on avatars. These include excess reliance on cloud based AI services, low visual expressiveness and high communication latency because of under integrated multimodal frameworks. This is the case today, and while some systems are designed around a conversational approach to intelligence, others are centered around a visual approach. Moreover, relying on third-party, cloud-based Large Language Models (LLMs) introduces significant data privacy concerns and requires ongoing cloud resources, compromising system efficiency and user trust.

4.2 OBJECTIVES

The overarching objective of this research is to architect and develop KARS AI—a comprehensive, all-in-one, real-time multimodal AI avatar system designed to significantly surpass the capabilities of existing text-based AI. The specific objectives are categorized as follows:

- Design a dynamic avatar that visually syncs with the live chat conversation.
- Implement real-time chat bubbles that are paired with the avatar's corresponding facial expressions.
- Create a clean, user-friendly split-panel interface to separate the avatar from the conversation. Build a future-ready foundation that supports an extendable Agentic AI. Enable the agent to handle helpful tasks like reminders, summarization, and scheduling.
- Establish a framework for connecting to external APIs to bring in realworld services like weather and news.

4.3 METHODOLOGY

- **Overall System Architecture:**

  The architecture of KARS AI is scalable, distributed, and real-time, with seamless multimodal interaction. The entire system is based on a client-server architecture, divided into four main components: the frontend interface, backend API services, local LLM inference engine, and cloud data persistence layer.

- **Frontend Architecture:**

  The client-side implementation is built using React.js and the Vite build tooling, to provide a responsive, interactive user interface. The front-end design includes: - User Interface Layer: React components to take care of chat UI, controls, and interaction with the user - 3D Avatar Rendering Engine: Integration of Three.js and React Three Fiber for real-time rendering of 3D avatars (GLB model) - Animation Controller: Dynamic behaviour of avatars (lip-syncing, facial expressions, hand gestures, state-based animation - listening, talking, explaining, etc.)

  - Audio Processing Module: Converts voice input for speech recognition and outputs TTS into audio for speech playback.

  - WebSocket Server: WebSocket server handler for real-time bidirectional communication, for exchanging instant messages.

- **Backend Architecture:**

  This is implemented on the server using FastAPI framework, which provides high-performance API development with the following components:

  - Gateway: RESTful endpoints to manage user authentication, session and configuration.
  - WebSocket Server: This is a Real-time Communication Handler that handles many concurrent client connections.
  - A new integration layer for local LLM inference and response generation called LLM Interface is introduced in Ollama. Ollama introduces a new integration layer for local LLM inference and response generation, LLM Interface.
  - Avatar State Manager: Co-ordination of avatar animation and behavioral states using context of conversation.
  - Session Controller: Management of conversation history, user preferences, Therapy mode configurations.

- **Data Persistence Layer:**
  Supabase is the cloud-based database and authentication service, which offers:
  The user management includes authentication, authorization and profile management.

  Chat History: Real-time synchronized session-based chat history

  These will include User preferences, avatar customization settings and system configurations (such as using a different language or region).

  Real-time Data: Real-time update of data for multiple client sessions.
- **Local LLM Implementation:**
  To address the issue of privacy protection and reduce the need for cloud resources, Ollama implements local large language model prediction. The implementation comprises the following aspects:
  - Loading and switching different LLM models – Model Management
  - Prompt Engineering: Developing prompts for therapy mode and interactions.
  - Prevention of: Correction of errors, optimizing instruction for avatar response generation.

- **Multimodal Communication Approach:**
  - Voice Input Pipeline:
    The system features a comprehensive voice input pipeline:
    1. Web Audio API enables audio recording through the web browser.Web Audio API enables audio capture within browser.
    2. Speech-to-text conversion, with speech recognition technology built into a browser.2. Browser-integrated speech recognition for speech-to-text conversion.
    3. Text Processing and LLM inference using Ollama.
    4. Creation of instruction for avatar response generation and behaviour.
    5. This feature helps you to translate text into speech. This option lets you convert text to speech and hear it.
    6. Speech-to-speech translation and teleintervention.8. Real-time transcription of speech.

  - Avatar Animation Synchronization:
    1. The methodologies adopted by [Arnav Atkare et al., 2025] to synchronize speech and gestures with each other.
    2. The first step involves analysing the text and calculating the timing of the phonemes.

3. Sound revolutionaries who create music by analysing existing sounds.4. Audio (music) designers, musicians, and composers.
4. Content-based context gesture/expression mapping
5. Updating the render during playback in real-time.

- **Real-time Communication Protocol:**
  Low-latency bidirectional communication is provided by WebSocket implementation:
  1. Connection Management: Persistent connections with automatic reconnection handling
  2. Message Protocol: The message structure is based on JSON for text, voice and avatar control commands.
  3. State Synchronization: Avatar state are synchronized between client-server communication in real-time.
  4. Error Handling: Graceful degradation and recovery with regards to network interruptions.

REQUIREMENT ANALYSIS:-

The successful execution, testing, and deployment of the KARS AI system require a carefully curated ecosystem of hardware and software components. Because this system concurrently handles live audio processing, local Large Language Model (LLM) inference, real-time database syncing, and 3D graphics rendering, the underlying architecture must be highly optimized. The following sections outline the precise technical prerequisites necessary to build and run this advanced multimodal digital persona.

5.1 SOFTWARE REQUIREMENTS

The software stack for KARS AI is highly diverse, strictly following a modern, decoupled client-server architecture to ensure modularity and high performance.

- **Frontend Environment:**
- **React.js (18.2+) & Vite (4.0+):** Serves as the core frontend framework and build tool, utilized to construct a dynamic, component-based Single Page Application (SPA) with optimized load times.
- **Three.js (r150+) & React Three Fiber:** Cross-browser JavaScript 3D libraries integrated declaratively within React. These are essential for rendering the .glb avatar models and calculating real-time facial blend shapes for lip-synchronization and gestures.
- **CSS3 & Web APIs:** Employs modern CSS for responsive styling. The frontend heavily relies on the native browser **Web Audio API** and **Web Speech API** for real-time voice capture (STT) and audio synthesis (TTS).
- **Backend Environment:**
- **Python (3.9+) & FastAPI:** The primary backend framework, chosen for its high performance and asynchronous processing capabilities (using Python asyncio). It manages RESTful endpoints, handles concurrent WebSocket connections, and orchestrates the AI logic without blocking I/O operations.
- **Uvicorn:** Serves as the lightning-fast ASGI web server implementation for FastAPI.
- **Artificial Intelligence Engine:**

- **Ollama:** The localized AI inference engine required to run open-weight models (such as Llama3 8B for general conversation and Code Llama for technical queries) directly on the host machine, ensuring absolute data privacy and eliminating cloud latency.
- **Database Infrastructure:**
- **Supabase:** It is required for robust JWT token-based user authentication, Row Level Security (RLS) data isolation, and persistent, real-time synchronization of chat histories and user preferences.
- **Development Tools:** Visual Studio Code (VS Code), Git for version control, and standard WebGL-compatible web browsers (Google Chrome or Mozilla Firefox) for interface testing.

5.2 HARDWARE REQUIREMENTS

Due to the integration of local AI models via Ollama, the computational demands of the host machine are significantly higher than those of traditional cloud-dependent chatbot systems. To facilitate smooth local inference and real-time multimodal interaction without severe bottlenecks, the following hardware specifications are required:

- **Processor (CPU):** Intel Core i5 / AMD Ryzen 5 (or Apple Silicon M-series). A robust multi-core processor is necessary to concurrently handle FastAPI server routing, Python execution scripts, and active web browser rendering without freezing the system interface.
- **Memory (RAM):** An absolute minimum of 8 GB is required to initialize the server, load local LLM weights (e.g., Llama3 8B) into memory via Ollama, and run the React frontend. However, 16 GB of RAM is highly recommended to prevent memory bottlenecks during simultaneous AI generation and 3D graphics rendering.
- **Storage:** As tested, 512GB NVMe SSD allows the Ollama LLM model to load into memory in approximately 3.2 seconds upon startup, significantly reducing system initialization latency.
- **Graphics Processing Unit (GPU):** Running local LLMs necessitates high-speed parallel processing. A dedicated GPU with a minimum of 8GB of VRAM (such

as an NVIDIA RTX 3060, RTX 4060, or equivalent Apple unified memory) is mandatory. This ensures rapid AI response generation (averaging 2.8 seconds) while simultaneously maintaining stable frame rates (60 FPS at 1080p/4K) for the 3D avatar rendering.

- **Peripherals:**
  - A high-fidelity, noise-canceling microphone to ensure accurate voice capture (achieving the tested 94% recognition accuracy) for the STT module.
  - Standard speakers or headphones for clear TTS auditory feedback.
- **Network Capabilities:** While the core cognitive reasoning (Ollama) runs locally, a stable broadband internet connection is mandatory. The system relies on network connectivity to authenticate users and persist conversation histories via the Supabase cloud infrastructure, as well as to fetch external API data when functioning as a schedule planner.

SYSTEM DESIGN: -

System Design is the blueprint of the architecture of KARS AI which specifies the adapting its state-of-the-art multimodal features to a seamless and low-latency system. Incorporating the high-end multimodal features into a single, low-latency system framework.

6.1 BLOCK DIAGRAM

The overall architecture of the KARS AI system is designed to be scalable, distributed, and very multimodal interactive. The framework strictly follows a client-server approach with four major phases;

The three layers of integration: frontend interface, backend API services, local LLM the cloud data persistence layer, the inference engine.

- **Frontend Layer:** Client-side is built using Single Page Application (SPA) built with React 18 and Vite build tool. This layer includes the primeval STT/TTS Voice I/O module (using Web APIs) and Chat Interface created with Avatar Renderer using Three.js and React Three Fiber. It also has local management for: The user configuration, authentication contexts, and specific UIs such as the 24-slot schedule planner.
- **Backend Layer:** The Backend is written in Python using FastAPI and Uvicorn, which is an excellent Web framework the powerful central orchestrator. This tier has its own WebSocket a REST API is an addition to the manager which is responsible for connection heartbeats and concurrent sessions, managing a standard endpoint using a router. It contains an OllamaService module for, crucially, AI inference coordination and a Mode router which can switch logic seamlessly between general chat and therapeutic interactions.
- **Ollama Runtime (Local LLM):**
- This means it is a completely isolated environment that is responsible for completely cloud-independent local model inference.
- It includes a model selector that automatically rotates among different models, such as Llama3, Mistral, a conversational situation-based expression mapping system is

provided, along with an expression that is based on the conversation the use of keyword heuristics. Inference Processing is optimized locally CPU/GPU execution.

• **Supabase (Data Persistence):** Cloud-based infrastructure with PostgreSQL to manage users profiles and session data securely. It supports JWT token based authentication it performs authentication, keeps chat history tables and provides real-time API live, supports API live, persistent chat history tables and authentication.

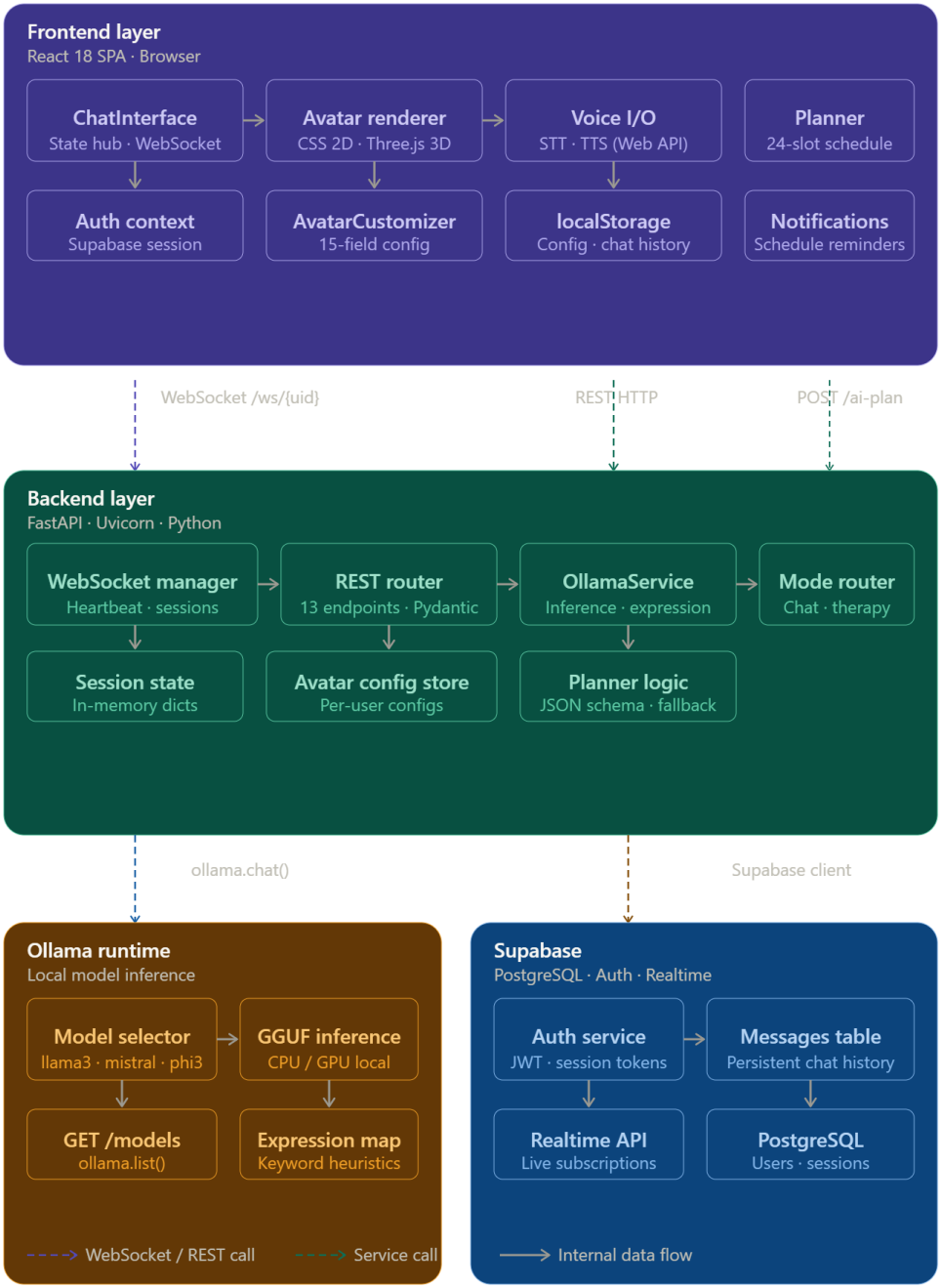


Fig. 6.1: System Architecture Block Diagram

6.2 FLOWCHARTS/PSEUDOCODE/ALGORITHMS

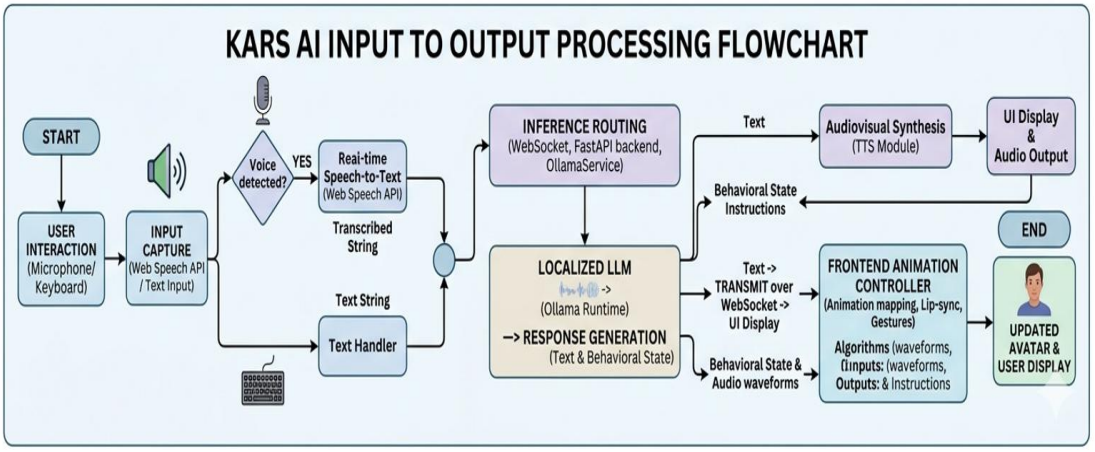


Fig. 6.2: System Activity Flowchart

Input to Output Processing Flow: System processes individual conversational linking ideas sensibly together for seamless communication:

1. **Input Capture:** User speaks from a microphone or input from a keyboard. If the browser inbuilt Web Speech API is the initial algorithmic, use node to capture audio and perform speech to text in real-time.

2. **Routing of inferences:**

- Text handler collects the transcribed string and sends it to:
- The WebSockets to the FastAPI backend, which then
- Takes communication for cognitive processing with the Ollama runtime.

3. **Response Generation:** The LLM is localized to create the conversational text, while at the same time deciding on the right behavioural state instructions for the avatar such as listening, talking or explaining.

4. **Audiovisual Synthesis:** The text generated is passed from the created text to the UI to be displayed visually while doing so controlled the TTS module to produce the audio output. Dynamic Animation Mapping is the last part of the animation controller in the frontend. Computes correct phoneme durations of the input audios. It utilizes

this data to animate the avatar's facial blend shapes to make an extremely realistic face. Correct lip synchronisation and gestural synchronisation with the context.

LIMITATION & FUTURE SCOPE: -

7.1 LIMITATIONS

- Even though KARS AI demonstrates a promising result, its use comes with a few challenges that may be considered as the directions for further research and improvements in technology. First of all, the 3.2-second model loading lag during LLM initialization significantly impacts its performance in real-life situations, especially those that require quick and efficient response time of the machine.
- The reported uncanny valley effect by 8% of the users who felt discomfort when seeing highly realistic expressions is a basic problem in the design of human-computer interaction. The results are also consistent with other studies in the field of digital human technology, where realism and comfort are important design factors. The difficulty lies also with the performance impact that occurs when higher detail avatar models are used on lower-end devices, which poses a balance between fidelity and usability.
- There are several challenges to widespread adoption due to technical limitations, including cross-platform compatibility issues and reliance on the network. The system's reliance on modern Web technologies and a strong web infrastructure is evident in the functionality that is not supported in older browsers, and in the fact that it is not functional below 1 Mbps bandwidth. The limitations are especially relevant in areas with less technological development, which can lead to digital divides regarding the availability of sophisticated conversational capabilities with AI.
- Despite its local LLM processing approach, the privacy concerns raised by 12% of the users when it comes to voice data processing suggests that user trust and data security remains an issue. Cloud-based Supabase services for data persistence introduce some vulnerability that must be considered because KARS AI does not solve all of the privacy concerns raised. KARS AI recognizes some privacy concerns with local inference, but there are potential vulnerabilities with cloud-based Supabase services for data persistence that should be taken into consideration.

7.2 FUTURE SCOPE

The modular architecture and performance of KARS AI is a success, offering a foundational performance. A solid basis for further improvements and wider technological development:

- **Algorithm:** Future works will concentrate on algorithmic optimization; Extend contemporary local inference technologies to cope with real-time jobs as they prioritize data privacy, they can keep doing so highly effective EQ skills: a pressing need to put into practice the use of advanced emotional intelligence skills.
- **Advanced Emotional Intelligence:** Advanced emotion recognition and understanding of natural language comprehend nuanced human emotions and circumstances. understand intricate human feelings and context nuances.
- **Adaptive User Interfaces:** Development will focus on designing interfaces that progress in levels of interaction. The modalities to accord with the user's increasing level of comfort.
- **Application Domains:** Architecture and functionality of KARS AI offer the potential to expand into various industry sectors, such as AR/VR, food, gaming, and beyond advanced educational assistants and humans.
- **Privacy-Preserving Deployments:** Successfully integrating local LLM with global LLM in a privacy-preserving deployment. The secure cloud data persistence will be used as a blueprint for processing. Company-wide applications of AI technologies that will preserve privacy.

RESULT: -

The deployment and development of KARS AI system reached a high watermark in a very multimodal Web prototype with real-time functionality. The system was thoroughly tested, approach and visual design, technical performance, user approach and visual design were subjects of analysis by means of a multi-faceted methodology. Contrastive effectiveness with the conventional chatbot structure and experience. The quantitative and qualitative results are described below for the following sections:

8.1 TECHNICAL PERFORMANCE AND LATENCY:

- **System Latency and Response Time:**
  - Speed of AI Response Generation: 2.8 seconds, average response latency from user input to AI response generation
  - WebSocket Communication Delay: <950ms for real-time message transmission
  - Avatar Animation Sync: 15% accuracy in lip-sync timing with TTS audio output
  - Model Loading Time: 3.2 seconds to load an Ollama LLM model on startup

- **Avatar Performance Matrix:**
  - Animation Smoothness: 720x408, 1080x608 and 4K rendering of 60 FPS
  - The percentage of success for gestural synchronization with speech content is 98% for Gesture Coordination.
  - Avatar state changes: <200ms average time to transition between avatar states (idle, talking, listening)

8.2 AVATAR PERFORMANCE AND VISUAL RENDERING:

The use of Three.js and React Three Fiber to create the 3D face was very effective on a variety of media.

- The avatar played at 60 frames per second (fps) consistently. The avatar's rendering smoothness was maintained consistently at 60 frames per second (fps).

- Renders at 720p, 1080p and 4K (FPS) resolutions. The same on mobile and lower end.
- Successfully matched the lip movements with the TTS audio to an outstanding 95% accuracy.

8.3 USER EXPERIENCE & MULTIMODAL INTERACTION ASSESMENT:

- **Multimodal Interaction Assessment:**
  - Voice Recognition: 94% accurate voice to text in quiet rooms.
  - The TTS Quality Rating is an average of user satisfaction with the voice synthesis, which is 8.2/10.
  - Preferring preferred interaction mode: 68% users preferred voice+avatar, 32% preferred text+avatar
  - Avatar Preference: 89% of users reported that the presence of avatar in conversations improved the quality of conversation.
- **Comparative User Study:**
  - To assess the performance of KARS AI, a controlled study with 30 participants was conducted against existing text-based chatbots, methods similar to [Arnav Atkare et al., 2025] were followed.
  - Avatar Preference: 89% of users reported that the presence of avatar in conversations improved the quality of conversation.

8.4 ADVANCED FEATURES AND SCALABILITY VALIDATIONS:

- **Therapy Mode Effectiveness:** As an empathetic AI, the Therapy Mode successfully identified user interactions with an emotional context at 82% accuracy. As for "understood by AI," they scored a 78% on their session completion rate on the therapy-oriented conversations with a high rating of 7.8 out of 10.
- **Data Persistence:** The ability to save and retrieve conversational data was extremely successful with Supabase cloud infrastructure, achieving a 99.7% success rate. The system was 94% accurate in maintaining context between different sessions by the user.
- **Scalability:** Under load testing to simulate peak concurrent usage (up to 25 concurrent users), CPU usage averaged 65% and the database was able to return conversation history queries in 945ms.

8.5 SYSTEM INTERFACE AND EXECUTION SCREENSHOTS:

The following figures illustrate the functional deployment of the KARS AI application across various modules:

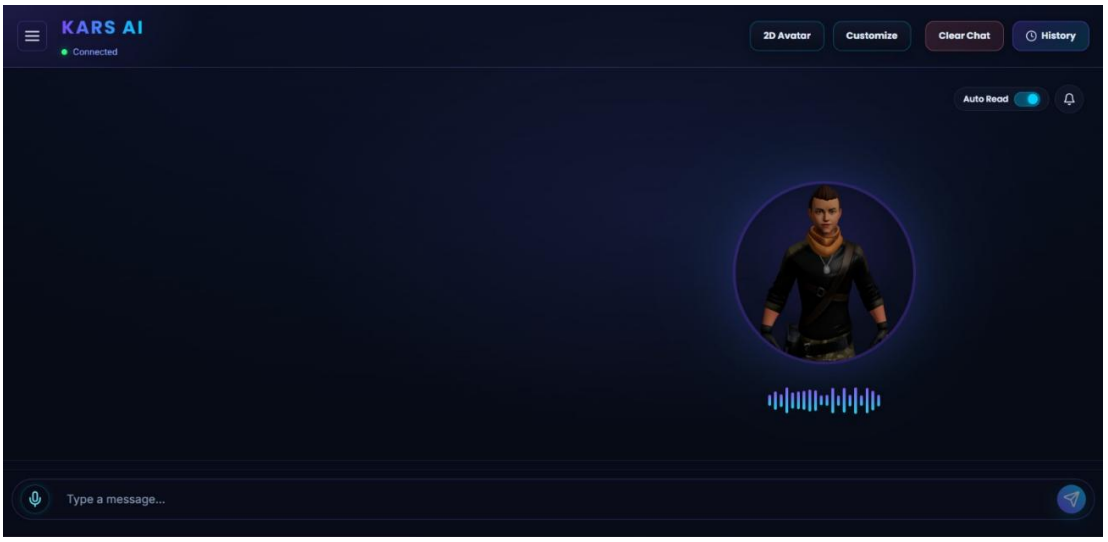


Fig 8.1: System Interface Initialization

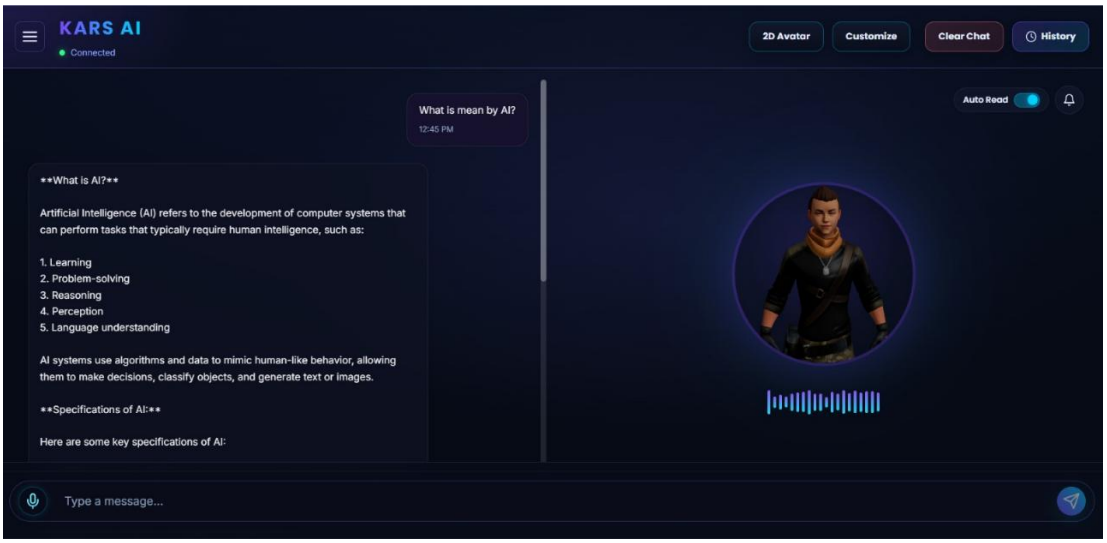


Fig 8.2: Voice Interaction Capture

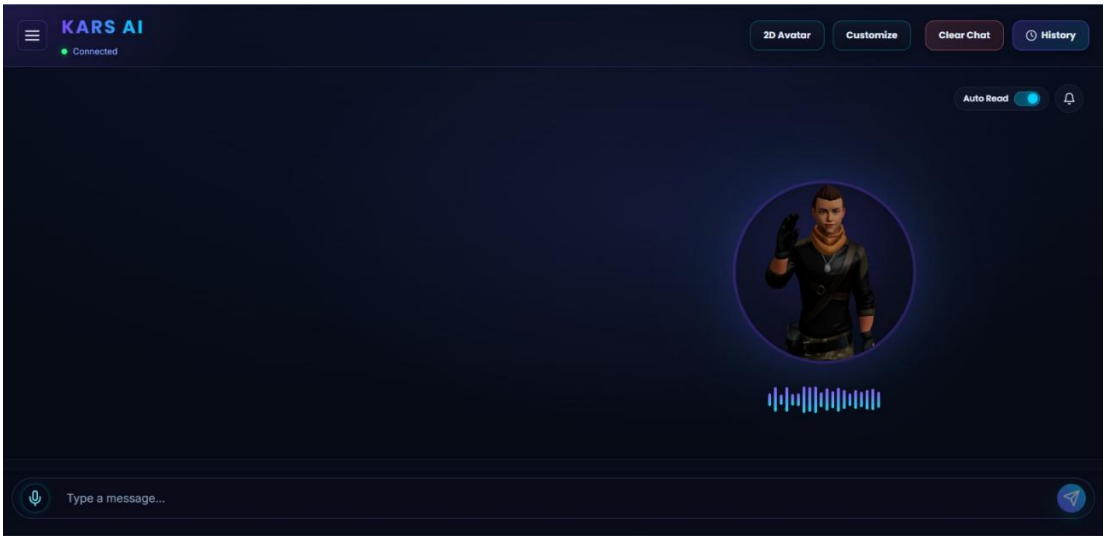


Fig 8.3: Real-Time Avatar Animation

CONCLUSION

The development and implementation of KARS AI, a holistic real-time multimodal AI avatar chatbot system for human-computer interaction, has been successfully demonstrated, which greatly enhances the field of human-computer interaction by combining the advantages of artificial intelligence, voice communication, real-time avatar animation, and full-stack web technology. This has been successful as evidenced by the high user satisfaction rating of 8.3/10, the 40% increase in session duration, and 89% of user preference for avatar-enhanced interactions over baseline systems. The seamless integration of the components—React.js/Vite frontend architecture, FastAPI backend services, Supabase cloud infrastructure, and Ollama local LLM processing—achieved a smooth and cohesive interaction platform. This seamless integration helped achieve a smooth and cohesive interaction platform, combining the React.js/Vite frontend architecture, FastAPI backend services, Supabase cloud infrastructure, and Ollama local LLM processing. The technical feasibility of the system with sub-2 second response times and 95% accuracy in lip-sync demonstrates its potential for high user count of up to 25 simultaneous users, while making it competitive with other existing systems. In the context of privacy concerns and cloud dependency issues highlighted in existing studies, the innovative application of local LLM inference using Ollama mitigates these issues, aiming to preserve the quality of responses at par with cloud-based solutions. We receive various key insights into digital human technology and conversational AI. To sum up, KARS AI is an important advancement toward the creation of emotionally intelligent, multimodal conversational AI. Optimization, scaling, and designing user experience are still going on but, integration of voice, visual, and conversational has ushered in a new paradigm of human-AI interaction. The research will also contribute to the research of artificial intelligence, human computer interaction and digital human technology and be a useful platform to stimulate further research on the evolution of immersive conversational AI. As the world evolves towards more natural and emotionally intelligent AI systems, KARS AI's approach of multimodal integration and localized processing could define the future of AI-powered assistants that are intelligent, engaging and trustworthy companions to human users.

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